Math 311 H — Recitation 4

Please hand in your completed work on Gradescope by **11:59 p.m. on Tuesday, September 29, 2026** with a complete and clearly written solution set.

1. (20 points) Recall that $\{z_n\}_{n\in\mathbb{N}}$ converges to $L \in \mathbb{R}$ if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $|z_n - L| < \varepsilon$ for every $n \geq N$.

   (a) Fix $L \in \mathbb{R}$. Decide which statements imply "$\{z_n\}_{n\in\mathbb{N}}$ does not converge to L." State and prove the implications that hold, and give counterexamples for those that fail.

   A. There exist $\varepsilon_0 > 0$ and $N \in \mathbb{N}$ such that $|z_n - L| \geq \varepsilon_0$ for every $n \geq N$.

   B. There exists $\varepsilon_0 > 0$ such that, for every $N \in \mathbb{N}$, there exists $n \geq N$ with $|z_n - L| \geq \varepsilon_0$.

   C. For every $N \in \mathbb{N}$, there exist $\varepsilon_N > 0$ and $n \geq N$ such that $|z_n - L| \geq \varepsilon_N$.

   Which statement is the exact negation of convergence to L? Does failure to converge to one particular L imply divergence?

   (b) Write out the definition of a divergent sequence and use it to prove that the sequence $a_n = 1 + (-1)^n$ diverges.

2. (25 points) Let $\{x_n\}_{n\in\mathbb{N}}$ be a real sequence such that $x_n^2 \to 4$.

   (a) Prove that $|x_n| \to 2$ by finding an estimate for $\big||x_n| - 2\big|$ in terms of $|x_n^2 - 4|$.

   (b) Must $\{x_n\}_{n\in\mathbb{N}}$ converge? Give a proof or a counterexample.

   (c) Suppose in addition that there exists $n_0 \in \mathbb{N}$ such that $x_n x_{n+1} \geq 0$ for every $n \geq n_0$. Prove that $\{x_n\}_{n\in\mathbb{N}}$ converges, and determine its possible limits.

3. (25 points) Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijection, and define $y_n = x_{\sigma(n)}$ for a real sequence $\{x_n\}_{n\in\mathbb{N}}$.

   (a) Prove that $x_n \to L$ if and only if $y_n \to L$.

   (b) Does the implication $x_n \to L \implies y_n \to L$ remain true if σ is onto but not necessarily one-to-one? Give a proof or an explicit counterexample, verifying that your map is onto.

4. (30 points) Let

$$a_n = \frac{(-1)^{n+1}}{n}, \qquad s_n = \sum_{j=1}^{n} a_j.$$

   For this question, *assume* that $s_n \to S$ for some $S > 0$. Following the rearrangement in Abbott, Section 2.1, define

$$b_{3k-2} = \frac{1}{4k-3}, \qquad b_{3k-1} = \frac{1}{4k-1}, \qquad b_{3k} = -\frac{1}{2k} \qquad \forall\, k \in \mathbb{N}$$

   and let $t_n = \sum_{j=1}^{n} b_j$. Thus the new order begins 1, 1/3, −1/2, 1/5, 1/7, −1/4, ...

   (a) Give an explicit bijection $\sigma : \mathbb{N} \to \mathbb{N}$ such that $b_n = a_{\sigma(n)}$. Verify that every original term is used exactly once.

   (b) Using only finite sums, prove that

$$t_{3m} = s_{4m} + \frac{1}{2} s_{2m} \qquad \forall\, m \in \mathbb{N}$$

   Hint: Separate the odd and even denominators in s_{4m} and s_{2m}.

   (c) Prove directly from the definition that $t_n \to 3S/2$.

   (d) Explain why the different limits of $\{s_n\}_{n\in\mathbb{N}}$ and $\{t_n\}_{n\in\mathbb{N}}$ do not contradict Question 3.